

Lab 3: Component Modelling & Architectural Pattern Selection

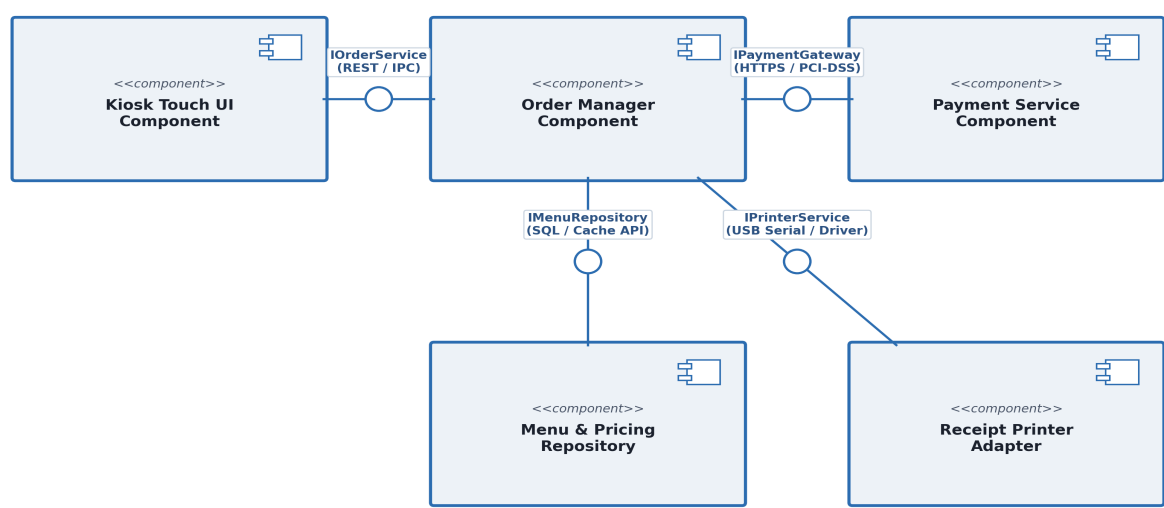
Scenario: Self-Service Coffee Kiosk System | Course: Software Engineering

1. System Components & Interface Specifications

Component Name	Stereotype	Role & Description	Interfaces Provided / Required
Kiosk Touch UI	«component»	Provides interactive touchscreen interface for coffee selection (Espresso, Americano, Latte) & sizes.	Required: IOrderService
Order Manager	«component»	Core business logic orchestrating order assembly, payment dispatch, printing, and inventory lookup.	Provided: IOrderService Required: IPaymentGateway, IPrinterService, IMenuRepository
Payment Service	«component»	Handles credit card authorization, tokenization, and transaction processing with external payment gateway.	Provided: IPaymentGateway
Receipt Printer Adapter	«component»	Hardware abstraction driver converting order receipts into ESC/POS thermal printing commands.	Provided: IPrinterService
Menu & Pricing Repository	«component»	Data storage layer managing coffee types, size pricing (Small/Large), and local in-memory menu cache.	Provided: IMenuRepository

2. UML Component Diagram

Self-Service Coffee Kiosk System - UML Component Diagram



3. Written Architectural Justification

Architecture Selection Statement:

"We chose a **Layered & Client-Server Component Architecture** for the Self-Service Coffee Kiosk System."

Scenario-Related Justification (Two Core Reasons):

- **Reason 1 (Hardware & UI Decoupling):** In a kiosk environment, hardware devices (touchscreen, credit card reader, thermal printer) are prone to physical wear or driver updates. Decoupling the *Kiosk Touch UI* and *Receipt Printer Adapter* behind explicit interfaces (*IPrinterService*, *IOrderService*) ensures hardware replacements or driver changes do not require altering core order processing logic.
- **Reason 2 (Modularity & Centralized Order Control):** Separating the *Order Manager* into a central controller enables seamless coordination between user selection, menu pricing lookups, payment processing, and receipt generation. If the café expands to online/mobile ordering in the future, the exact same *Order Manager* and backend services can be reused without redesigning the system.

Security Advantage:

- **Isolated Payment Processing & PCI-DSS Compliance:** Credit card processing is completely encapsulated inside the dedicated *Payment Service Component*. The *Kiosk Touch UI* and *Order Manager* never store, log, or persist sensitive Primary Account Numbers (PAN) or CVV details. Payment requests are transmitted over encrypted TLS channels directly to external payment gateways, isolating PCI-DSS audit scope strictly to the Payment Service component.

Performance Benefit:

- **In-Memory Caching & Asynchronous Receipt Printing:** The *Menu & Pricing Repository* maintains an in-memory local cache of coffee types and size pricing, enabling sub-10ms UI rendering when customers browse beverages. Furthermore, receipt generation in the *Receipt Printer Adapter* operates asynchronously on a background thread, allowing the touchscreen UI to immediately display an order completion screen without waiting for physical paper printing.